PE Definition Document

Description: A highly efficient, instruction-set driven Processing Element (PE) for accelerating quantized neural network inference.

PE Operation:

The PE operates as an instruction-driven hardware accelerator designed to offload computationally intensive neural network inference tasks. Controlled by a Chip Select signal, the PE transitions between a low-leakage dormant state and an active execution mode. The PE operates on quantized signed integers, which serve as efficient approximations of the original floating-point number model parameters. Upon enablement, the internal controller decodes a 5-bit operational code (Op-Code) to manage data routing and arithmetic precision. The input valid signal acts as a sampling enable, instructing the PE to read the op-code and data buses, causing activity in the following clock cycles.

Following the input validity signal, two registers ingest signed 32-bit packed integer operands (inputs and weights). The PE can dynamically reconfigure its multipliers and adders to support either high-fidelity INT8 or high-throughput INT4 operations. The computational pipeline executes a Multiply and Accumulate (MAC) sequence into a 64-bit register (bias can be added through a separate instruction), followed by a rescale of the 64-bit signed accumulator value to the nearest 4/8-bit signed integer. A hardware-based activation function finalizes 4/8-bit inference result. Activation output is driven to the output bus, which is synchronized with a valid-output signal to ensure seamless integration with host. The unit is fed a System Reset signal as well, which clears all internal registers when active.

Op-Code:

Op-Code	Mnemonic	Full Name	Description
00000	NOP	No Operation / Sleep	No registers change value and no calculations being made.
00001	RST_ACC	Reset Accumulator	Clears the 64-bit signed accumulator register.
00010	MAC4	Multiply and Accumulate in INT4	PE treats the 32-bit inputs as eight 4-bit signed integers. See full details in PE operation section.
00011	MAC8	Multiply and Accumulate in INT8	PE treats the 32-bit inputs as four 8-bit signed integers. See full details in PE operation section.
00100	ADD_BIAS	Add Bias	Adds the 32-bit bias to the accumulator from operand_A/B bus (t.b.d).
00101	RESCALE	Rescale	Rescales the 64-bit signed accumulator into an INT8 activation input. See full details in PE operation section.
00110	ACTIVATE	Activation Function	Applies activation function on rescaled activation input and passes it to 8-bit output register. See full details in PE operation section.
00111-11111	RSV	Reserved	Reserved for other instructions that would be needed later in production.

Instructions:

MAC8 vs MAC4 Operation: the MAC instructions perform a dot-product on operand_A and operand_B inputs, Input mapping is flexible; the user may assign weights and NN inputs to either operand bus:

$$\sum_{i=1}^{N} input_i * weight_i \rightarrow accumulator$$

- The MAC8 instruction configures the datapath for INT8 precision; 32-bit operands are interpreted as four packed, signed 8-bit integers, which are multiplied in parallel, then added into a 64-bit accumulator.

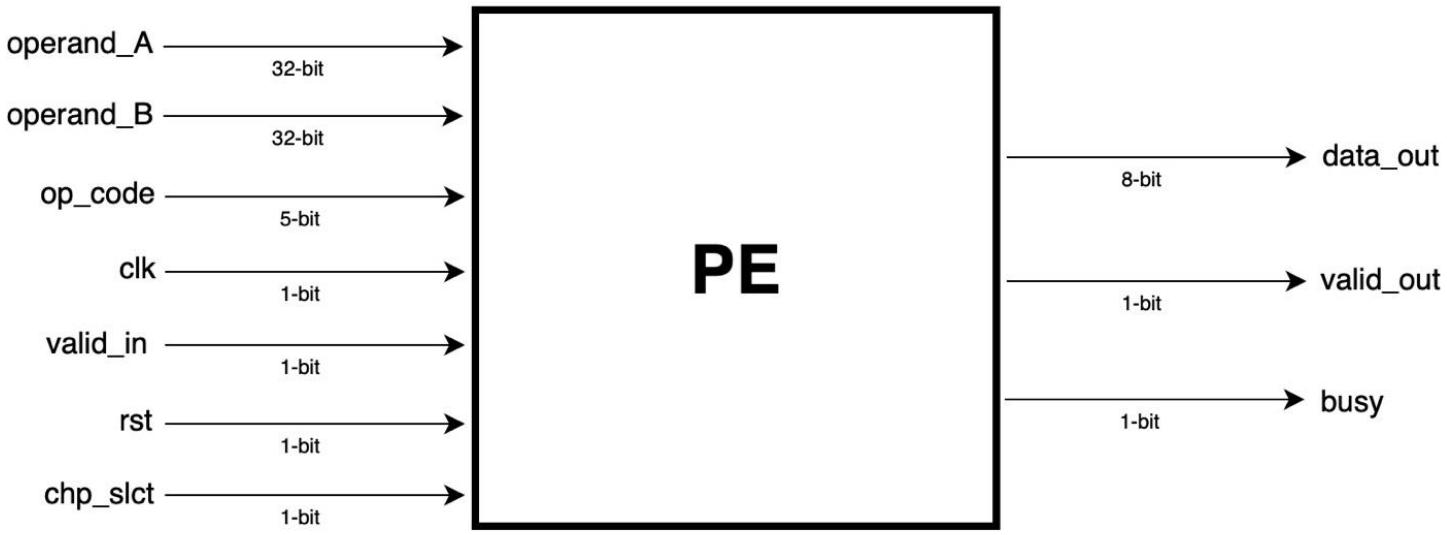
- The MAC4 instruction configures the datapath for INT4 precision; 32-bit operands are interpreted as eight packed, signed 4-bit integers, which are multiplied in parallel as well (high-throughput mode), then added to an accumulator.

Reset Accumulator: This instruction clears the accumulator, ensuring data independence between consecutive inference cycles. No operands are used for the execution of this task. Accumulator state at power-up is to be disclosed at a later stage of production.

Rescale: The rescale operation multiplies the 64-bit signed accumulator value with a 64-bit scaling factor, then an unsigned 6-bit shift factor (with a sign extension to 64-bit) will round the quantized 64-bit value into the nearest 4/8-bit signed integer. The result will be placed on the 8-bit activation input register. The 64-bit scaling factor and the unsigned 6-bit shift values are dependent on the precision mode; exact values are to be disclosed at a later stage of production.

Activation Function: The final stage of the computational pipeline is the Activation unit, responsible for applying transformations to the accumulated data. The unit maps the normalized 4/8-bit output from the Rescale stage into a corresponding activation value based on the user's desired activation function (Linear, ReLU, Leaky ReLU, Tanh). The Tanh activation utilizes a LUT; activation output will be undefined if the table has not been initialized.

I/O:



Power-Up State: For a cleared state at power-up, the user must activate the System Reset signal. The power-up state of the PE is to be determined at a later stage of production.

System Reset: A synchronic, rising edge activated System Reset input signal clears all internal registers and initializes the datapath to the default MAC4 precision mode.

All signals are sampled at the clock's rising edge:

Signal Name	Signal Type	Width (bit number)	Direction	Function	User Note
`operand_A`	Data	32	Input	NN Inputs / Weights / Bias	**Packed Integer.** Contains four signed INT8s or eight INT4s. See instruction section.
`operand_B`	Data	32	Input	NN Inputs / Weights / Bias	**Packed Integer.** Contains four signed INT8s or eight INT4s. See instruction section.
`op_code`	Control	5	Input	Operational Code	See op-code table and instruction section.
`clk`	Global	1	Input	System Clock	**Rising edge active.**
`valid_in`	Control	1	Input	Input Validity	Signals inputs are valid. See PE operation section.
`rst`	Global	1	Input	System Reset	Resets registers and controllers to a defined initial state. See above for further details.
`chp_slct`	Control	1	Input	Chip Select	When inactive – operands and op-code aren't sampled; Operations started before inactivity resume.
`data_out`	Data	8	Output	NN Output	Contains final 8-bit inference result.
`valid_out`	Global	1	Output	Output Validity	Activated when data_out contains a valid result.
`busy`	Control	1	Output	Status Flag	When active - PE can't receive instructions and data; when inactive - ready to receive new inputs from user (on the following clock cycle).